DELIVERABLE 3 | GROUP CAPSTONE PROJECT

ERP Solution Design: TasteBuds Restaurant Chain

Target ERP System: Odoo ERP Enterprise	**Scope:** Multi-Company / Multi-Branch Architecture
Core Lifecycle Focus: P2P, O2C & Ticket-to-Table	**Document Version:** 1.0 (Final)

1. System Architecture & Module Map

The ERP solution is designed around five core Odoo applications, interconnected to create a seamless real-time data loop across branches and central HQ.

Odoo Application	Target Process	Key Operational Capabilities & Justification
Point of Sale (POS)	O2C & Ticket-to-Table	Table layout management, order routing to KDS, bill splitting, and fast payment collection. Powers FOH operations.
Purchase	Procure-to-Pay (P2P)	Requisitions, RFQs, vendor blanket agreements, and automated PO approval threshold workflows ($500 limit).
Inventory	P2P & Ticket-to-Table	Multi-location warehouses, FIFO automated valuation, scrap/spoilage tracking, and automated BOM ingredient deduction.
Invoicing & Accounting	P2P & O2C	3-way vendor bill matching, POS daily journal aggregation, automated general ledger posting, and bank reconciliation.
Employees (HR)	System Security	Role-Based Access Control (RBAC) matrix defining explicit operational permissions per role across branches.

2. Technical Data Architecture & Integration Logic

A. Bill of Materials (BOM) & Inventory Deduction

Finished goods sold at POS (e.g., *TasteBuds Special Burger*) are configured as consumable products linked directly to a Manufacturing / POS BOM. When an order is validated:

POS Order Validation (1x Special Burger)
↓ *Triggers Automated FIFO Deduction* ↓
Raw Beef Patties: -0.20 kg | **Burger Buns:** -1.0 unit | **Fresh Tomatoes:** -0.05 kg
Updated in Real-Time within Branch Main Kitchen Inventory

B. Kitchen Display System (KDS) Routing Rules

Orders entered at table-side POS tablets automatically route based on product categories:

- **Grill & Fryer Category:** Routes to *Back-of-House Kitchen Screen 1*.
- **Beverages & Desserts Category:** Routes to *Bar Display Screen 2*.
- **KDS Stage Lifecycle:** `To Do` → `In Preparation` → `Ready` → `Served`.

3. Role-Based Access Control (RBAC) Matrix

To preserve data integrity across branches, user access is strictly regulated:

User Role	Purchase App	Inventory App	POS & KDS	Accounting
Line Cook / Expediter	--	--	KDS View Only	--
Waitstaff / Cashier	--	--	POS User	--
Head Chef / Branch Mgr	User (Requisitions)	User (Stock/Scrap)	POS User	--
Central Purchasing Officer	Administrator	User	--	--
Company Accountant	--	--	--	Administrator

4. Cross-Functional Business Process Integration

1. Procure-to-Pay (P2P) Lifecycle

Stock drops below minimum threshold → System auto-generates RFQ → Central Purchaser consolidates orders & approves PO → Kitchen receiving validates receipt (or scraps damaged items) → Accountant performs 3-Way Match (PO vs Receipt vs Bill) → Vendor Payment Registered.

2. Ticket-to-Table (Kitchen Fulfillment) Lifecycle

Waitstaff inputs order at tablet POS → Order routes dynamically to Kitchen KDS screens → Line cook toggles stage to "In Preparation" → Expediter performs QC and marks "Ready" → Waitstaff notified → Food served to table.

3. Order-to-Cash (O2C) Lifecycle

Guest finishes meal → Waitstaff prints preliminary bill → Customer pays via Cash/Card/Split → POS validates receipt → Shift close aggregated journal entries post automatically to Accounting General Ledger.

5. Executive KPI Dashboards Configuration

- **CEO Dashboard:** Real-time consolidated multi-branch sales, top-selling dishes, overall gross profit margin %, and sales channel breakdown.
- **Sales Dashboard:** Hourly POS sales velocity, table turnover rates, average check size per table, and payment method distribution.
- **Inventory Dashboard:** Stock valuation (FIFO), real-time raw ingredient levels, wastage/spoilage log, and automated reorder triggers.
- **Finance Dashboard:** Daily POS revenue reconciliation, accounts payable aging, 3-way match audit logs, and bank account synchronization status.